Module 1: Network + Network Fundamentals and building networks

Ethernet Standard and Specification

Task 1:

Task 2:

	10BASE-T	100BASE-TX	1000BASE-T
Speed	10 Mbps	100 Mbps	1000 Mbps (1 Gbps)
Common Name	Ethernet	Fast Ethernet	Gigabit Ethernet
Cable Type	Cat3 or higher (UTP)	Cat5 or higher (UTP)	Cat5e or higher (UTP)
Max Cable length	100 meters	100 meters	100 meters
Connector type	RJ-45	RJ-45	RJ-45

Task 3:

Use a crossover cable.

In a straight-through cable, pin 1 (transmit) on one end connects to pin 1 (transmit) on the other end — that works fine when one device is a "sending" type and the other is a "receiving" type (like a computer to a switch), because the switch internally crosses the signal for you. But when connecting **two identical devices directly** — like two laptops — both NICs try to transmit on the same pins and listen on the same pins, so nothing gets received. A crossover cable swaps the transmit and receive pairs internally, so laptop A's transmit pins line up with laptop B's receive pins, and vice versa, allowing direct communication.

Task 4:

Three home/college devices and their likely Ethernet standard

1. **Wi-Fi router**— likely supports **1000BASE-T (Gigabit Ethernet)**, since most modern routers need to handle fiber broadband speeds well above 100 Mbps.
2. **Desktop computer or laptop in a college computer lab** — likely uses **100BASE-TX or 1000BASE-T**, depending on the age of the machine's network card; newer labs are typically Gigabit.
3. **Network printer shared across an office/college department** — often still runs on **100BASE-TX**, since printers don't need high bandwidth and many older/cheaper printer NICs are only rated for Fast Ethernet.